

Lloyds BCB Data Collector

Dissertation Project - Complete README
Lloyds Banking Group · Commercial Banking Division · BCB Sector Intelligence

5 free APIs

Zero

8 sectors

~40 features

Data sources

API keys needed

BCB sectors covered

Cleaned dataset

1. Project Overview

This toolkit collects, processes, and merges publicly available data to identify commercial banking opportunity signals for Lloyds BCB (Business and Commercial Banking) target sectors. It supports the dissertation research question:

"Can external public data (company filings + media coverage) be used to predict business needs and inform commercial banking strategies?"

1.1 The 8 BCB Target Sectors

Lloyds specified the following sectors as priority targets for BCB engagement:

Sector	SIC Code Range	Key Characteristics
Manufacturing	10000 – 33999	Production, output signals
Public Sector, Education & Charities	84000 – 88999	Government-adjacent
Healthcare	86000 – 87999	NHS, private health
Technology, Legal & Professional	62000 – 74999	High growth potential
Agriculture	1000 – 3999	Seasonal, rural businesses
Real Estate	68000 – 68999	Commercial property
Wholesale & Retail	45000 – 47999	Consumer-facing
Fast Growth & Emerging	Cross-cutting	Identified by financial signals

2. Data Sources

All five sources are free to use. No credit card or paid subscription is required.

2.1 Companies House Bulk Download

What it is

The UK's official company register. Monthly snapshot of every registered UK company — name, number, SIC code, accounts filing dates, mortgages, status.

- Company name, registration number, address
- SIC codes (industry classification)
- Incorporation date and company status
- Accounts filing dates and overdue status

- Mortgage and charge information
- Account category (micro / small / full)

https://download.companieshouse.gov.uk/en_output.html

Free bulk download, no registration needed. File size: ~700MB ZIP.

2.2 Companies House Filing API

What it is

REST API providing filing history, officer (director) appointments and resignations per company. Free with optional key registration at developer.company-information.service.gov.uk

- Director appointment and resignation counts
- Late filing detection (days after deadline)
- Historical compliance pattern

Note: Basic access works without a key but is rate-limited to ~600 requests per 10 minutes.

2.3 GDELT (Global Database of Events, Language and Tone)

What it is

Global news monitoring platform tracking media in 100+ languages. Provides article-level sentiment scores, event data, and volume trends. Completely free, no key needed.

- Average news tone per company (positive = growth, negative = stress)
- Percentage of negative articles (risk signal)
- Sector-level sentiment trends over 12 months
- Volume trend (is the sector getting more/less coverage?)

<https://api.gdeltproject.org/api/v2/doc/doc>

2.4 OpenCorporates

What it is

Global company database covering multiple jurisdictions. Useful for identifying UK-operating companies registered abroad. Free tier, no key required.

- International parent company identification
- Cross-border subsidiary mapping
- Company type and industry cross-reference

2.5 Data Source Summary Table

Source	What you get	API Key?	Cost	File
Companies House Bulk	All UK companies	No	Free	<code>fetch_bulk.py</code>
Companies House API	Filings, directors	Optional	Free	<code>fetch_filings.py</code>
GDELT	News sentiment	No	Free	<code>fetch_gdelt.py</code>
OpenCorporates	Cross-border data	No	Free	<code>fetch_opencorporates.py</code>

3. Project File Structure

`lloyds_data_collector/`

`run_all.py`

← START HERE: runs everything

requirements.txt	← Python dependencies
README.md	← Quick reference
Data_Cleaning.ipynb	← Step 7: clean and prepare final dataset
companies_house/	
fetch_bulk.py	← Step 1: download bulk data
filter_sectors.py	← Step 2: filter for BCB sectors
fetch_filings.py	← Step 3: director & filing data
gdelt/	
fetch_gdelt.py	← Step 4: news sentiment
news/	
fetch_opencorporates.py	← Step 5: cross-border data
utils/	
sic_codes.py	← SIC code → sector mapping
merge_all.py	← Step 6: join all sources & create labels
output/	
companies_bcb_filtered.csv	
filings_and_directors.csv	
gdelt_sentiment.csv	
gdelt_sector_trends.csv	
opencorporates_enriched.csv	
final_model_dataset.csv	← 3.1M rows, 110 cols, corrected labels
cleaned_companies_house_dataset.csv	← FINAL: 3.1M rows, 40 cols, ready for modelling

4. Setup & Installation

4.1 Requirements

- Python 3.8 or higher
- Internet connection (for downloads and API calls)
- ~2GB free disk space (for bulk download)
- 15–30 minutes run time depending on connection speed

4.2 Install Dependencies

```
pip install -r requirements.txt
```

The requirements.txt installs:

- requests — HTTP calls to all APIs
- pandas — data processing and CSV handling
- numpy — numerical operations
- tqdm — progress bars
- rapidfuzz — fuzzy company name matching
- textblob — additional sentiment analysis

4.3 Quick Start (Run Everything)

```
python run_all.py
```

This runs all 6 steps in order. If a step fails it will ask whether to continue. You can also run individual steps manually (see Section 5).

5. Script Reference

5.1 fetch_bulk.py

Purpose	Downloads the Companies House monthly bulk snapshot (all UK companies)
Input	None — downloads directly from Companies House website
Output	output/raw/companies_house_bulk/*.csv
Run time	5–15 minutes depending on internet speed (~700MB download)
Key behaviour	Skips re-download if ZIP already exists. Delete the ZIP to force re-download.

```
python companies_house/fetch_bulk.py
```

5.2 filter_sectors.py

Purpose	Reads bulk CSV, filters for 8 BCB sectors by SIC code, engineers features
Input	output/raw/companies_house_bulk/*.csv
Output	output/companies_bcb_filtered.csv
Run time	5–10 minutes (processes ~5M rows in chunks)
Key features created	company_age_years, is_active, is_dormant, accounts_overdue_days, fast_growth_proxy, and 10 more

```
python companies_house/filter_sectors.py
```

5.3 fetch_filings.py

Purpose	Fetches filing history and director changes per company from Companies House API
Input	output/companies_bcb_filtered.csv (samples 200 companies per sector)
Output	output/filings_and_directors.csv
Run time	20–40 minutes (rate-limited to be polite)
API Key	Optional — works without one, slower rate limit applies

```
python companies_house/fetch_filings.py
```

5.4 fetch_gdelt.py

Purpose	Queries GDELT for news sentiment at company level and sector level
Input	output/companies_bcb_filtered.csv
Output	output/gdelt_sentiment.csv + output/gdelt_sector_trends.csv
Run time	30–60 minutes (50 companies per sector × 8 sectors)
API Key	Not required — GDELT is permanently free

```
python gdelt/fetch_gdelt.py
```

5.5 fetch_opencorporates.py

Purpose	Enriches companies with cross-border and international data from OpenCorporates
Input	output/companies_bcb_filtered.csv
Output	output/opencorporates_enriched.csv
Run time	10–20 minutes (300 company sample)
Rate limit	~500 requests/day on free tier — script pauses automatically if hit

```
python news/fetch_opencorporates.py
```

5.6 merge_all.py

Purpose	Joins all 5 data sources on CompanyNumber, creates proxy labels
Input	All 5 output CSVs from previous steps
Output	output/final_model_dataset.csv + output/final_model_dataset_summary.txt
Run time	Under 1 minute
Missing sources	If a source CSV is missing it is skipped — partial merges still work

```
python utils/merge_all.py
```

5.7 Data_Cleaning.ipynb

Purpose	Cleans final_model_dataset.csv — removes high-missing cols, imputes values, drops constants
Input	output/final_model_dataset.csv
Output	cleaned_companies_house_dataset.csv (3,145,434 rows × 40 cols)
Run time	5–15 minutes (loads full 1.7GB file into memory)
Note	Use Jupyter or run via: <code>python -c "import nbformat; ..."</code> or extract cells manually

6. Final Dataset Columns

The cleaned_companies_house_dataset.csv contains 40 columns grouped into categories. The raw final_model_dataset.csv has 110 columns before cleaning.

6.1 Identity Columns (drop before training)

Column	Type	Meaning
CompanyNumber	text	8-digit unique ID — use as join key only
CompanyName	text	Full registered company name
bcb_sector	text	One of the 8 BCB target sectors — encode as dummies for model
SICCode.SicText_1	text	Raw SIC code text from Companies House

6.2 Feature Columns (X — feed into model)

Column	Type	What it means
company_age_years	float	Years since incorporation. Young = growth potential.
is_active	0/1	1 = currently trading. Only active companies are prospects.
is_dormant	0/1	1 = shell company with no activity. Not a prospect.
is_micro	0/1	1 = turnover under £632k. Very early stage business.
is_small	0/1	1 = turnover under £10.2m. Core BCB target size.
is_full_accounts	0/1	1 = large enough to file full accounts. Bankable size.
has_changed_name	0/1	1 = traded under a previous name. Possible rebrand signal.
accounts_overdue_days	int	Days past accounts deadline. 0 = healthy. 100+ = risk.
conf_stmt_overdue_days	int	Days overdue on confirmation statement.

accounts_ever_late	0/1	1 = has filed accounts late at least once.
has_late_filing_history	0/1	1 = pattern of repeated late filings.
num_mortgages_outstanding	int	Active charges registered. Already borrowing = may need more.
has_active_mortgages	0/1	Binary version of above.
director_turnover_signal	0/1	1 = high director churn in past year. Risk/change signal.
director_resignations_1yr	int	Raw count of directors who resigned in 12 months.
avg_tone	float	Average GDELT news sentiment (-10 to +10). NULL for most companies — see note.
pct_negative	float	Percentage of articles with strongly negative tone.
sector_avg_tone	float	Sector-wide average news tone for context.
sector_volume_trend	float	Is news coverage of this sector growing or shrinking?

6.3 Proxy Label Columns (y — prediction targets)

Important: these are proxy labels, not ground truth

These labels were constructed from observable public signals because direct Lloyds product uptake data is not available. State this clearly in your dissertation methodology.

Column	Values	Meaning	Distribution
label_growth_opportunity	0 or 1	PRIMARY TARGET. 1 = young active company, not dormant, no overdue accounts. Positive GDELT sentiment (if available) used as bonus override.	56.67% positive
label_risk_signal	0 or 1	1 = ever filed accounts late, or confirmation statement overdue >180 days, or has late filing history, or high director turnover. Negative GDELT sentiment used as bonus override.	6.58% positive
label_lending_need_proxy	0 or 1	1 = has active mortgages + active + not flagged as risk. Hardest to defend as a label — use with caution.	12.90% positive

7. Modelling Guide

7.1 What You Are Predicting

Your model predicts label_growth_opportunity (0 or 1) for each company. In plain English:

"Does this company show the profile of a growing business that would benefit from Lloyds commercial banking services?"

7.2 Train/Test Split Code

Data Leakage Warning

The proxy labels were built from specific feature columns. If you include those same columns as model inputs, the model will trivially reconstruct the label formula and report near-perfect accuracy that is meaningless. For label_growth_opportunity: exclude company_age_years, is_active, is_dormant, accounts_overdue_days. For label_risk_signal: exclude accounts_ever_late, conf_stmt_overdue_days, has_late_filing_history, director_turnover_signal. For label_lending_need_proxy: exclude has_active_mortgages and label_risk_signal. Use the remaining columns as your feature set so the model learns genuine patterns.

```
import pandas as pd
```

```

from sklearn.model_selection import train_test_split

df = pd.read_csv('cleaned_companies_house_dataset.csv')

# Exclude label-defining columns to prevent data leakage
FEATURES = [
    'is_micro', 'is_small', 'is_full_accounts',
    'has_changed_name',
    'num_mortgages_outstanding', 'has_active_mortgages',
    'director_resignations_1yr',
    'avg_tone', 'pct_negative',
    'sector_avg_tone', 'sector_volume_trend'
]
TARGET = 'label_growth_opportunity'

X = df[FEATURES].fillna(0)
y = df[TARGET]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

```

7.3 Model Training (XGBoost)

```

from xgboost import XGBClassifier
from sklearn.metrics import classification_report

model = XGBClassifier(random_state=42, eval_metric='logloss')
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print(classification_report(y_test, y_pred))

```

7.4 Generating the Final Ranked Output

```

df_out = df[['CompanyNumber', 'CompanyName', 'bcb_sector'] + FEATURES].copy()
df_out['predicted_label'] = model.predict(X)
df_out['predicted_prob'] = model.predict_proba(X)[:, 1]
df_out = df_out.sort_values('predicted_prob', ascending=False)
df_out.to_csv('output/predictions.csv', index=False)

```

The predicted_prob column (0.0 to 1.0) is the key output. Sort by this to get a ranked list of the strongest opportunities per sector for Lloyds relationship managers.

8. Dissertation Methodology Notes

8.1 How to Describe the Proxy Labels

Suggested dissertation wording

"In the absence of ground-truth commercial banking uptake data, binary proxy labels were constructed from publicly observable company characteristics (filing compliance, company age, trading status, mortgage activity, director turnover) as surrogate measures of growth opportunity, risk, and lending need, following established practice in commercial intelligence research. News sentiment data from GDELT was incorporated as a supplementary signal where available, though coverage of small private UK companies was found to be negligible in practice, so structured Companies House signals form the primary basis for all three labels."

8.2 Key Limitations to Discuss

- Proxy labels are not ground truth — they approximate likely need, not confirmed product uptake

- GDELT returned zero news articles for all sampled companies in this dataset — small private UK firms rarely appear in global news. Labels are therefore entirely derived from structured Companies House data; avg_tone is NULL for ~100% of the population and fills to 0 in model training
- Companies House bulk data is a snapshot — some data may be slightly outdated
- OpenCorporates free tier has a ~500 request/day limit — only a sample is enriched
- Fast Growth sector is identified by signals, not SIC code — acknowledge this approximation
- Label-defining features must be excluded from the model feature set to avoid data leakage (see Section 7.2)

8.3 Key Strengths to Highlight

- Entirely replicable using free, public data — no proprietary sources
- Covers all 8 Lloyds BCB target sectors systematically
- Combines structured (Companies House) and unstructured (GDELT news) data as specified in the brief
- Labels rebuilt from population-wide signals ensure non-degenerate distributions across 3.1M companies
- Produces an actionable ranked output that relationship managers could use directly
- Feature importance from XGBoost provides explainability for each prediction

9. Troubleshooting

Problem	Fix
Bulk download URL not found	Go to download.companieshouse.gov.uk and copy the latest URL manually into <code>fetch_bulk.py</code>
GDELT returns empty results	Company name too specific or no news coverage — expected for small private UK firms. NULL avg_tone is filled with 0 in merge step and does not affect label creation.
All label values are 0 (constant)	Labels now depend on structured columns (is_active, accounts_ever_late, etc.), not GDELT. Run <code>merge_all.py</code> again to regenerate with the updated label logic.
OpenCorporates 429 error	Rate limit hit. Script waits 60 seconds automatically. Reduce <code>SAMPLE_SIZE</code> if needed.
<code>final_model_dataset.csv</code> is missing columns	Run <code>merge_all.py</code> after ALL other scripts complete. Missing source files are skipped.
XGBoost import error	Run: <code>pip install xgboost</code>
Memory error on bulk CSV	The file is ~700MB. Close other applications or reduce chunksize in <code>filter_sectors.py</code>
Model accuracy is suspiciously near 100%	Data leakage — you have included label-defining columns in your FEATURES. See Section 7.2 for the correct feature list.